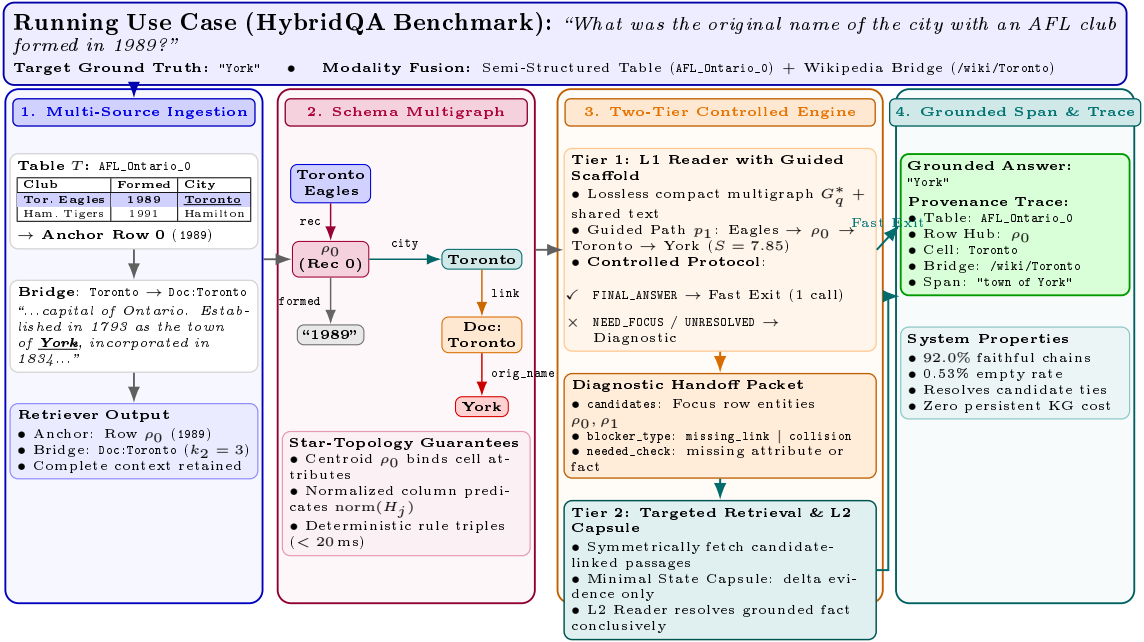

Running Use Case (HybridQA Benchmark): "What was the original name of the city with an AFL club formed in 1989?"
Target Ground Truth: "York" • Modality Fusion: Semi-Structured Table (AFL_Ontario_0) + Wikipedia Bridge (/wiki/Toronto)
1. Multi-Source Ingestion
Table T: AFL_Ontario_0
Club | Formed | City
Tor. Eagles | 1989 | Toronto
Ham. Tigers | 1991 | Hamilton
→ Anchor Row 0 (1989)
Bridge: Toronto → Doc:Toronto
"...capital of Ontario. Established in 1793 as the town of York, incorporated in 1834..."
Retriever Output
• Anchor: Row $\rho_0$ (1989)
• Bridge: Doc:Toronto ($k_2 = 3$)
• Complete context retained
2. Schema Multigraph
Toronto Eagles
rec
$\rho_0$ (Rec 0)
city
Toronto
link
formed
"1989"
Doc: Toronto
orig_name
York
Star-Topology Guarantees
• Centroid $\rho_0$ binds cell attributes
• Normalized column predicates norm($H_j$)
• Deterministic rule triples (< 20 ms)
3. Two-Tier Controlled Engine
Tier 1: L1 Reader with Guided Scaffold
• Lossless compact multigraph $G_q^*$ + shared text
• Guided Path $p_1$: Eagles → $\rho_0$ → Toronto → York ($S = 7.85$)
• Controlled Protocol:
✓ FINAL_ANSWER → Fast Exit (1 call)
× NEED_FOCUS / UNRESOLVED → Diagnostic
Fast Exit
Diagnostic Handoff Packet
• candidates: Focus row entities $\rho_0, \rho_1$
• blocker_type: missing_link | collision
• needed_check: missing attribute or fact
Tier 2: Targeted Retrieval & L2 Capsule
• Symmetrically fetch candidate-linked passages
• Minimal State Capsule: delta evidence only
• L2 Reader resolves grounded fact conclusively
4. Grounded Span & Trace
Grounded Answer:
"York"
Provenance Trace:
• Table: AFL_Ontario_0
• Row Hub: $\rho_0$
• Cell: Toronto
• Bridge: /wiki/Toronto
• Span: "town of York"
System Properties
• 92.0% faithful chains
• 0.53% empty rate
• Resolves candidate ties
• Zero persistent KG cost